Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Beginathon

Selected Domain

Domain 2: Orbyte Smart SME Challenge (Alpha Red Solutions)

Selected SDG Track (1–17)

SDG 8 (main) · also SDG 12

Project / Solution Name

Just Right: AI bake planner for small cafés and bakeries

Name (Leader)

Derek Chay Wen Hong

Email & Phone (Leader)

pixel4470@gmail.com · 011-1246 2899

University Affiliation(s)

Multimedia University (MMU), Cyberjaya

Number of Team Members

2 (Derek Chay Wen Hong, Tan Jun Hao)

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small cafés and bakeries in Malaysia decide every day how many croissants, loaves and muffins to bake, and most decide from memory. Baking too much means unsold food goes in the bin at closing. Baking too little means items sell out by lunch and full-price sales walk out of the door. Because nobody records leftovers or sell-out times, the same mistakes repeat week after week.

Those affected are owners and managers of micro and small food businesses, their staff, and the environment. For the business it is lost margin and wasted ingredients; for the country it adds to a large food-waste problem: SWCorp figures reported in June 2026 show Malaysians discarding about 6,000 tonnes of edible food every day. We visited three bakeries: Bakefield, an independent bakery, estimates its daily bake by rule of thumb and discounts unsold bread at closing; the chains we saw rely on policy instead (La Boheme bases quantities on past sales and discounts daily; Hokkaido bakes again when items sell out and returns unsold stock to its factory).

The two mistakes do not cost the same. Throwing away a croissant loses its ingredient cost (about RM 2), while running out loses the profit on a sale (about RM 6), so ‘zero waste’ is the wrong goal. Small shops have no analyst and no forecasting tools to find the right amount for each item.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Just Right is a bake-planning module for the Orbyte business hub that tells the manager how many of each item to bake tomorrow.

At closing, staff tap the leftovers into a Telegram bot (about 30 seconds), and roughly when anything sold out. An AI demand forecast then predicts how many of each item customers will want tomorrow. It is a machine-learning model (scikit-learn linear regression, one per item) that learns each shop's own patterns from past sales, the day of the week and the weather forecast from the Open-Meteo API. It retrains after every closing count, so it keeps learning, and it checks itself by comparing its past predictions with what really sold. On our simulated data it misses by about 2.4 pieces per item per day, against 3.2 for a plain 28-day average.

The prediction then goes into a newsvendor rule: high-margin items are planned slightly above expected demand because running out costs more than a leftover. The manager receives one Telegram message with each number and a plain reason (e.g. 'rain -7, Sunday +3'), approves or edits it in one tap, and the baker receives the final list. Every action is logged; a web dashboard shows the plan, AI accuracy, waste in RM and the activity log.

Workshop skills used: GenAI-assisted development, database design (SQLite), REST API design and cloud deployment on Replit.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Messaging: Telegram Bot API (pyTelegramBotAPI) for staff, manager and baker.

Backend / API: Python, Flask (JSON API + dashboard), APScheduler (daily timers).

AI forecast: scikit-learn linear regression per item (trained on CPU), NumPy.

Decision rule: newsvendor critical-ratio rule with a ±15% daily change cap.

External data: Open-Meteo weather API (history + forecast, cached).

Database: SQLite (daily sales, plans, activity log).

Frontend: manager dashboard in HTML + React (CDN) + Chart.js, served by Flask.

Hosting: Replit Deployments (Reserved VM, Asia); GitHub for version control.

Orbyte: simulated with synthetic items, costs, staff roles and POS sales.

Block diagram: see Appendix A (last page).

Sponsor-provided GenAI API or Cloud Platform: Yes. Replit (sponsor-provided Replit Core) hosts the whole application as an always-on deployment and stores the Telegram token and user IDs in Replit Secrets. RunPod was not used because the model is small and trains on CPU.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we visited three bakeries (Bakefield, La Boheme, Hokkaido) to learn how quantities are decided and what happens to leftovers: the independent shop guesses, while chains have policies and factory back-up. **Define:** the burden is daily overproduction and sell-outs, not selling leftovers. **Ideate:** our first idea, a clearance marketplace, was dropped after critique and mentor feedback because it needed a consumer audience and did not fit Orbyte as a business hub. **Prototype:** we built the smallest loop that works end to end: closing count, forecast, plan, approval, bake list and log. **Test:** unit tests cover the forecast and planner, and the model is back-tested day by day on historical data.

For usability we use Telegram, which staff already know, so there is no training and only one message per day for the manager. The prototype is already deployed on Replit with a live dashboard and bot. For Phase 2 we will refine it with real shop feedback and mentor input.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 12.3:** by 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains. **Target 8.2:** achieve higher levels of economic productivity through diversification, technological upgrading and innovation.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**National:** about 6,000 tonnes of edible food discarded per day in Malaysia in 2025, up from about 5,000 tonnes in 2024 (SWCorp data, reported by New Straits Times, 15 June 2026). **Shop level:** none of the 3 bakeries we visited records a forecast; the independent bakery (Bakefield) estimates by rule of thumb and discounts unsold bread at closing. **Our simulated bakery:** baking the usual fixed batch wastes about RM 340 per week across 3 items.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	1) **Weekly food waste per shop** in RM and pieces (leftovers × cost), measured together with the number of sell-outs so waste is not cut by under-baking. 2) **Forecast error:** average pieces off per item per day, compared with a plain 28-day average.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**20% lower weekly food waste** (RM) than the usual-batch baseline on our simulated test, with no increase in sell-outs. **Forecast error at least 25% lower** than a plain average (currently 2.4 vs 3.2 pieces). Based on simulated data; to be re-checked with real shop data.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☒ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable. All project code, design and data were created during the hackathon period (from 19 October 2026).

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Flask 3.0.3, APScheduler 3.10.4, python-dotenv 1.0.1	code (web server, timers, config)	BSD-3-Clause, MIT
pyTelegramBotAPI 4.21.0	code (Telegram bot client)	GPL-2.0
scikit-learn, NumPy, requests	code (ML library; model trained by us; HTTP)	BSD-3-Clause, Apache-2.0
React, Chart.js 4.4.1 (CDN)	code (dashboard UI and charts)	MIT
Open-Meteo weather data	dataset (public weather API)	CC BY 4.0

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable. No team member has an employment, internship or consulting agreement.

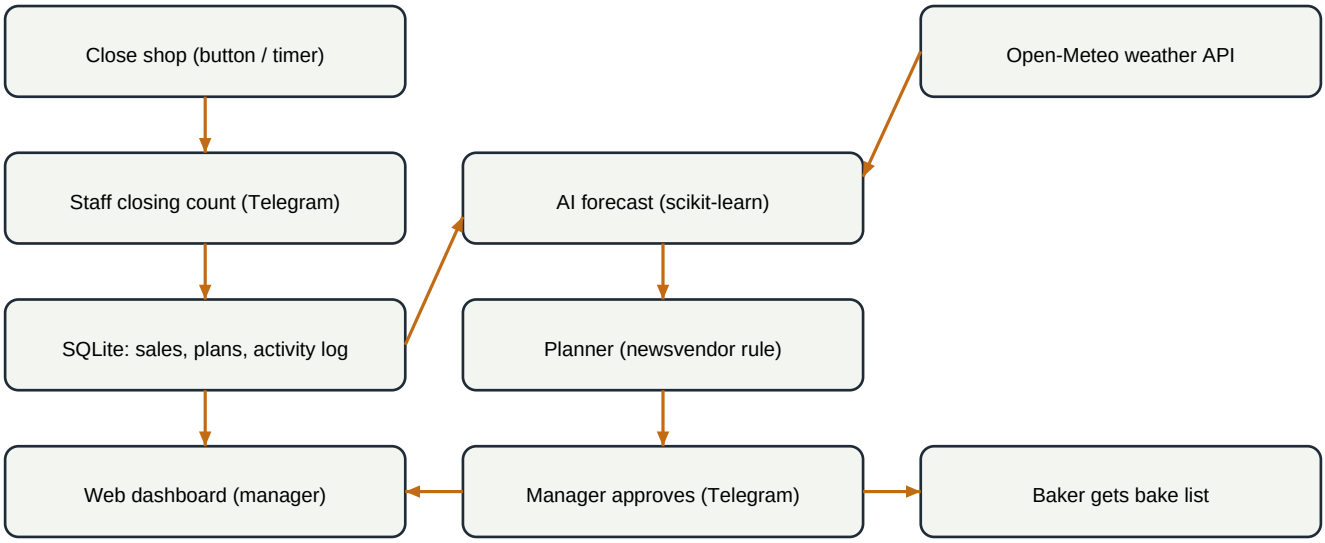
7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]
Derek Chay Wen Hong, Team Leader

[Date]
24 October 2026

Appendix A: System Block Diagram



Everything runs as one Python process on Replit (Reserved VM). Orbyte data (sales, costs, roles) is simulated.